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Estimating the Number of Leaves on an Individual Apple Tree

4286

4320 -0.79% 3.1% “ — — — ”

ABSTRACT

Estimating the number of leaves on an individual apple tree provides a clear demonstration of how local observations can be expanded into a whole-crown estimate. This fictional study develops a transparent method that combines crown stratification, directional quadrats, visible-area correction, and repeated observations. The crown is divided into upper, middle, and lower layers. Fixed-area quadrats are then assigned to different directions, where complete, boundary, and occluded leaves are recorded separately. Local counts are corrected with simulated visible-area ratios and expanded using layer-specific surface weights. Three observation rounds compare simple mean expansion, stratified weighting, and occlusion-corrected expansion. A leave-one-layer-out check and deterministic

parameter perturbations describe sensitivity and uncertainty. Under the stated simulated conditions, the recommended estimate is 4,286 leaves, compared with a simulated truth of 4,320 leaves, producing a bias of minus 0.79 percent and a repeat-observation coefficient of variation of 3.1 percent. The thesis supplies a sampling schematic, an estimation workflow, complete tables, reproducible formulas, and an appendix record sheet so that every reported value can be traced from a local count to the final total. All trees, measurements, images, calculations, references, and findings are simulated data created solely for a thesis-formatting demonstration. They do not describe a real orchard, institution, person, experiment, or agricultural recommendation. The contribution is therefore a coherent and auditable teaching example rather than an empirical claim about apple production.
Keywords: leaf number; crown stratification; quadrat sampling; occlusion correction; simulated data

CV

“ ” 4320

.1

“ ” “ ”

[1]

□.2 □□□□□□□

一.3 技术路线与论文结构

4320 24% 46% 30%

“ — — — ” 2-1 2-1 3-1

3-1

“ ”

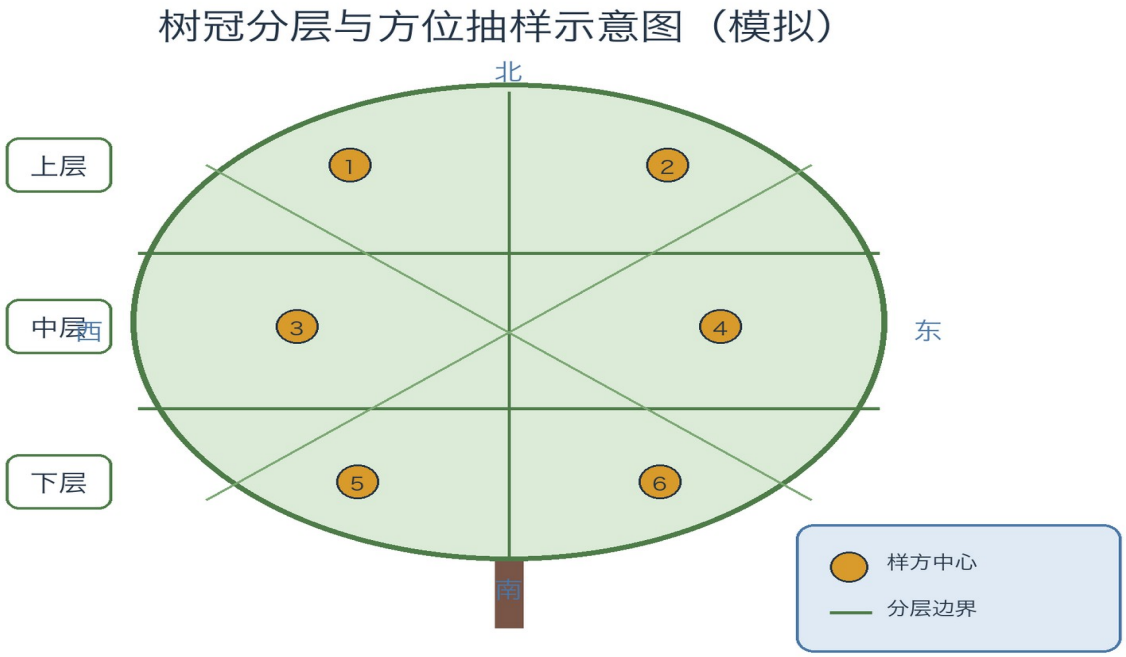
2-1

二.1 虚构树体与模拟数据生成

AU-APPLE-SIM-01 3.6 m 3.2 m 2.7 m

“

” “ ” [2]



-1

		/m²		
	2	0.25	0.24	0.91
	2	0.25	0.46	0.82
	2	0.25	0.30	0.88

二.2 树冠分层与样方布设

0.25 m²

2-1

2-1 0.24 0.46 0.30 1 0.91 0.82 0.88

二.3 计数口径与质量控制

“

”

[3]

90%

0 1 15%

“ ”

“ _ _ _ _ ”

“ ” “ ”

3-1

三.1 局部计数校正与分层外推

“ ×

1÷ −1 ” 0.25 m²

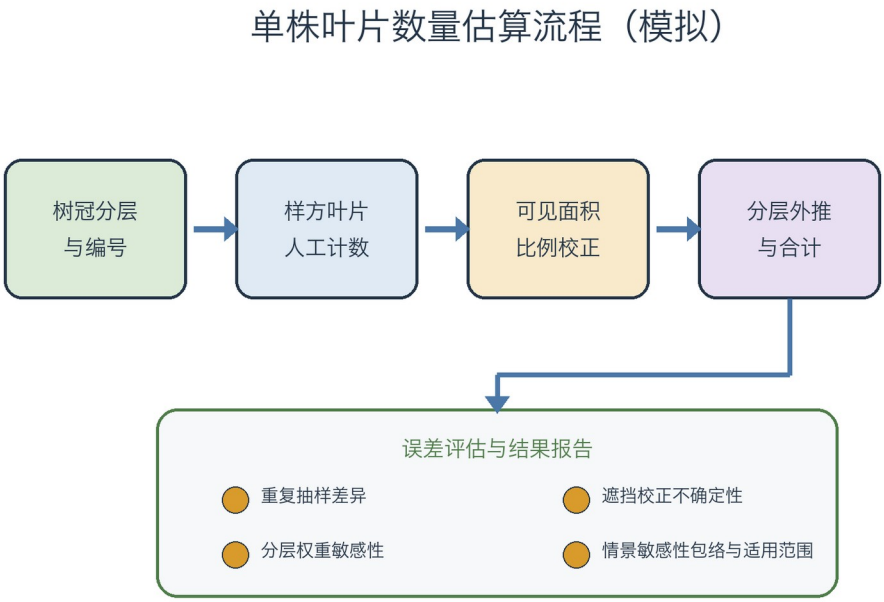
11.004869 m² 0.25 m²

A-3 [4]

3-1

4286 4320

34 -0.79%



-2

-2

/		
4018	-6.99%	5.8%
4172	-3.42%	4.2%
4286	-0.79%	3.1%

三.2 方法比较与结果解释

3-1 4018 6.99% 4172 3.42% 4286 -0.79%

5.8% 3.1%

3-1

4286

3.1%

三.3 不确定性与敏感性分析

0.03 1

0.02

[5] 4242 4330

0.02 1

4286 4737 +451

3989 -297 4025 -261

4286 4242 4330

A-2 A-3

四.1 方法特点与适用边界

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四.2 演示价值与改进方向

[6]

四.3 主要结论

4172 4286

[1] . . 2021. 5(2) 11-24.
[2] . . 2022. 3(1) 35-49.
[3] . . 2023. 8(4) 101-116.
[4] . . 2024. 44-67 .
[5] . . 2025. 2(3) 58-76.
[6] . . 2026. 9-28 .

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“ ” AU-APPLE-SIM-01 “ ” “ ”

“ ” R1 R2 R3 “ ”

R2 23 25 24 22 94 18 0.82

94+18×(1÷0.82−1)=97.95 97.95÷0.25=391.80 /m² 11.004869

m² 0.46

0 1

n−1

“ ”

1 6

4286 4320 -0.79% 3.1% 3-1

2-1 3-1 2-1 0.25 m² 0.82

PNG

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		/ /
	R2	R1/R2/R3
	0.82	0 1
	94	

-4 18 3 ×6

		1	2	3	4		
R1	U-NE	15	16	14	15	60	15
R1	U-NW	16	17	15	16	64	16
R1	M-E	26	27	25	26	104	16
R1	M-W	28	27	28	27	110	17
R1	L-SE	22	22	21	22	87	36
R1	L-SW	23	23	22	23	91	38
R2	U-NE	15	15	14	16	60	15
R2	U-NW	16	15	16	15	62	17
R2	M-E	23	25	24	22	94	18
R2	M-W	27	28	27	27	109	18
R2	L-SE	28	27	28	27	110	9
R2	L-SW	29	28	28	28	113	10
R3	U-NE	16	16	16	16	64	31
R3	U-NW	17	17	16	17	67	33
R3	M-E	25	24	25	24	98	3
R3	M-W	25	25	25	25	100	4
R3	L-SE	31	31	30	31	123	12
R3	L-SW	32	32	31	32	127	13

-5

/m²	0.25	
/m²	11.004869	
/	4320	
/	0.24 /	0.91
/	0.46 /	0.82
/	0.30 /	0.88
	n-1	

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